



MD. FAIAAD RAHMAN

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📈 CARRER OBJECTIVE

Electrical and Electronic Engineering graduate with strong foundations in circuit design, control systems, and renewable energy technologies. Interested in contributing to the development of efficient and sustainable engineering solutions through analytical problem-solving and innovative design. Seeking an opportunity in a challenging engineering environment that values technical excellence, continuous learning, and impactful technological development.

🎓 EDUCATIONAL BACKGROUND

UNIVERSITY - BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY 2023-2025

MASTER OF SCIENCE (ENGG.) IN ELECTRICAL AND ELECTRONIC ENGINEERING

- CGPA: 3.5/4.0 (Theory coursework completed)
- Result Published: 24th February, 2025

UNIVERSITY - SHAHJALAL UNIVERSITY OF SCIENCE AND TECHNOLOGY 2017-2022

BACHELOR OF SCIENCE (ENGG.) IN ELECTRICAL AND ELECTRONIC ENGINEERING

- CGPA: 3.3/4.0
- Result Published: 2nd March, 2023

COLLEGE - BIRSHRESTHA NOOR MOHAMMAD PUBLIC COLLEGE 2014-2016

HIGHER SECONDARY CERTIFICATE (HSC) GROUP: SCIENCE

- GPA: 5.0/5.0

HIGH SCHOOL - ANNADA GOVT. HIGH SCHOOL 2009-2014

SECONDARY SCHOOL CERTIFICATE (SSC) GROUP: SCIENCE

- GPA: 5.0/5.0

📁 PART-TIME WORKING EXPERIENCE

ACADEMIC CONTENT WRITING Nov, 2022 - Sept, 2023

- Livingstone Research, London, UK
- Status: Part-time
- Position: Academic Content Writer

UNDERGRADUATE RESEARCH ASSISTANT June, 2020 - April, 2022

- Department of Electrical and Electronic Engineering, Shahjalal University of Science and Technology, Sylhet.
- Supervisor: Tahmid Aziz Chowdhury, Showmik Singha

⚙️ SKILLS

PROGRAMMING LANGUAGES C | Python | MATLAB | Arduino (Microcontroller)

SOFTWARES Microsoft Powerpoint | Microsoft Word | Microsoft Excel | LaTeX | Proteus | Simulink
Cadence Virtuoso | Adobe Photoshop | Adobe Illustrator | AutoCAD | Ansys Lumerical
Blender

LANGUAGES *Native:* Bengali *Professional:* English

INDUSTRIAL TRAINING

1. **BANGLADESH POWER DEVELOPMENT BOARD (BPDB)** 15-09-2021
 - **Focused Area:** Power generation of 225 MW combined cycle powerplant (Gas+Steam), HRSG, Damper, Protection system of the powerplant (different type relays, surge arrestor), Auxiliary transformer, CT, PT, GCB.
2. **WORKSHOP ON INTRODUCTION TO ROBOTICS – ROBO-SUST** 22/4/17-06/05/17
 - **Focused Area:** Use of micro-controllers (Arduino), Basic knowledge of robotics, Built a line follower & android phone controlled two-wheeled robot.

THESIS RESEARCH AREAS

Solar Cell Materials | Design & Modeling of Solar Cell Device | Renewable Energy Systems

EXTRA CURRICULAR ACTIVITIES

Former General Member of **RoboSUST**

Former General Member of **IEEE SUST Student Branch**

PROJECTS

University Funded Project

"Intra-departmental document transferring system using smart quadcopter." DOI: [10.13140/RG.2.2.27952.28164](https://doi.org/10.13140/RG.2.2.27952.28164)

Academic Projects

1. Schematic and layout design of a 4-Bit incrementor/decrementor using Cadence Virtuoso.
2. Design and implementation of a DC–DC boost converter (24V to 40V) with minimized ripple.
3. Electromagnetic radiation detection in a mobile communication system using Arduino.
4. Vehicle accident alert system using accelerometer, gyroscope, fire & smoke sensor.
5. Wheelchair control by head motion using accelerometer for physically handicapped people.
6. 2D Residential floor design using AutoCAD.
7. Implementation of a movie recommendation system using ML models.

PUBLICATIONS

Journal Publications

1. "Performance improvement of three terminal heterojunction bipolar transistor based hybrid solar cell using nano-rods," *Solar Energy, Elsevier*, vol. 240, pp. 1–12, 2022. DOI: [10.1016/j.solener.2022.05.006](https://doi.org/10.1016/j.solener.2022.05.006)
2. "Hierarchically structured and thermally robust perovskite solar cells with band-engineered double-hole layers," *Solar Energy Materials and Solar Cells, Elsevier*, vol. 299, p. 114201. DOI: [10.1016/j.solmat.2026.114201](https://doi.org/10.1016/j.solmat.2026.114201)
3. "Unveiling architectural and optoelectronic synergies in lead-free perovskite/perovskite/kesterite triple-junction monolithic tandem solar cells", *Submitted to Solar Energy Materials and Solar Cells, Elsevier*. DOI: [10.48550/arXiv.2511.06059](https://doi.org/10.48550/arXiv.2511.06059)

Conference Proceedings

1. "Performance analysis of $\text{MgF}_2\text{--Si}_3\text{N}_4$ and $\text{MgF}_2\text{--Ta}_2\text{O}_5$ double-layer anti-reflection coating on heterojunction bipolar transistor solar cell," *Lecture Notes in Electrical Engineering*, vol. 616, Springer. DOI: [10.1007/978-981-16-1978-6_25](https://doi.org/10.1007/978-981-16-1978-6_25)
2. Optoelectronic analysis of lead free FASnI_3 single junction solar cell incorporating Cu:NiOx HTL," — Accepted in 2nd IEEE QPAIN (2026).

AWARDS & SCHOLARSHIPS

🏆 Awards:

1. '**Best Paper Award for Track-2: Renewable Energy Systems**' — 'Symposium on Power Electronics & Renewable Energy Systems Control (PERESC-2020)', IIT Bhubaneswar, Odisha, India.
Event Date: 04th-05th December, 2020.
2. '**2nd runner-up in Semiconductor Poster (Digital Display)**' — BEAR Summit and National Semiconductor Symposium, 2025, Dhaka, Bangladesh.
Event Date: 16th-18th July, 2025.

🎓 Scholarships:

1. Govt. Junior School Board Scholarship (2011)
2. Govt. Secondary School Board Scholarship (2014)

+ REFERENCES

Reference-1:

[Dr. Ahmed Zubair](#)

🎓 Professor,

🏢 Dept. of Electrical & Electronic Engineering, BUET

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Reference-2:

[Tahmid Aziz Chowdhury](#)

🎓 Assistant Professor,

🏢 Dept. of Electrical & Electronic Engineering, SUST.

✉ tahmid-eee@sust.edu

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Reference-3:

[Showmik Singha](#)

🎓 Assistant Professor,

🏢 Dept. of Electrical & Electronic Engineering, SUST.

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